

Session 1.4: Scale Dynamics & Winner-Takes-All

Market tipping, HHI concentration analysis, switching frictions, and Uber China vs. Didi.

PART 1 (LECTURE)
60 Minutes

PART 2 (LAB PRACTICE)
60 Minutes

PART 3 (EXECUTIVE MEMO)
60 Minutes

STANDARD
Audited SEC Filings

Learning Objectives & Executive Overview

Clear conceptual grounding for undergraduate and graduate business students.

CORE COMPETENCIES

Goals

What You Will Master Today

- **Theoretical Mastery:** Evaluate seminal literature and analytical frameworks governing platform competition.
- **Empirical Rigor:** Reconcile management strategic claims with legally audited statutory filings.
- **Data Modeling:** Execute spreadsheet calculations with explicit coordinates and formula logic.
- **Executive Formulation:** Deliver structured board-level recommendations resolving complex trade-offs.

WINNER-TAKES-ALL DYNAMICS

Vocabulary

Key Topical Foundations

- **Arthur (1989)::** Path dependency, lock-in, and historical tipping.
- **Eisenmann Criteria::** High multi-homing costs & strong cross-side effects.
- **HHI Metric::** Herfindahl-Hirschman Index industry concentration.
- **Micro-Examples::** Search engines (monopoly), mobile OS, ride apps.

Theoretical Foundations: Winner-Takes-All Dynamics

Rigorous analytical frameworks and literature benchmarks.

THEORETICAL ARCHITECTURE

Literature

Seminal Framework Insights

- **Arthur (1989)::** Path dependency, lock-in, and historical tipping.
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STRATEGIC APPLICATION

Analysis

Business Model Mechanics

- **Billion-Dollar War::** Subsidy battle across China (2014-2016).
- **Homogeneous Demand::** Riders & drivers switching for pennies.
- **Failure to Tip::** Scale alone failed to produce monopoly pricing power.
- **Strategic Truce::** ^{Uber} Uber sold China ops to Didi for 18.8% equity.

STRATEGIC TAKEAWAY

Market tipping, HHI concentration analysis, switching frictions, and Uber China vs. Didi.

Seminal Literature & Economic Mechanics

Governing economic dynamics, externalities, and platform governance models.

ECONOMIC EXTERNALITY

Model

Network Externalities & Market Tipping

- **Positive Feedback Loops:** Adoption utility accelerates quadratically as the complementary user base expands.
- **Two-Sided Subsidization:** Balancing prices between money-side and subsidy-side to catalyze cross-network participation.
- **Multi-Homing Frictions:** Users and suppliers incurring setup and cognitive costs when operating across competing platforms.
- **Governance Standards:** Programmatic rules and interface contracts mitigating opportunistic market behavior.

STRATEGIC LEVER

Ecosystem Dynamics

Platform Architecture Mechanics

- **Arthur (1989)::** Path dependency, lock-in, and historical tipping.
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Strategic Showdown: Incumbent Architecture vs. Platform Ecosystem

Contrasting capital-intensive linear controls with scalable ecosystem orchestration.

LINEAR INCUMBENT MODEL

Physical Pipeline

Controlled Capacity & Heavy CapEx

- **Proprietary Asset Ownership:** Firms fund physical plant, equipment, and inventories directly on balance sheet.
- **Gatekeeper Curation:** Central corporate management manually selects products, vendors, and services.
- **Supply Chain Bottlenecks:** Expansion speed is strictly constrained by capital expenditure budgets and construction lead time.
- **Linear Economics:** Revenue growth requires proportional expansion in headcount and physical fixed assets.

PLATFORM ECOSYSTEM MODEL

Software Orchestrator

Algorithmic Matching & Asset-Light Scale

- **Externalized Production:** Community participants supply physical capital, labor, and distributed inventory.
- **Algorithmic Governance:** Two-way reputation systems and automated matching replace physical gatekeepers.
- **Zero Marginal Cost Expansion:** Adding new marketplace participants requires lines of code rather than land acquisition.
- **Exponential Leverage:** Transaction volume accelerates rapidly while corporate headcount remains tightly bounded.

CORE STRATEGIC DISPARITY

Pipelines compete by optimizing internal operations and erecting proprietary barriers. Platforms compete by orchestrating external ecosystems and minimizing friction across bilateral exchanges.

Case Showdown: Uber China vs. Didi

Historical case evidence and multi-year corporate trajectory confrontation.

CORPORATE DYNAMICS

Timeline & Strategy

Uber

Strategic Positioning

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ANALYTICAL FINDINGS

Empirical Evidence

Operational & Financial Results

- *Market Share Data::* Search, Mobile OS, and US Ride-Hailing shares.
- *HHI Calculation::* Sum of squared market shares (SUM(Share^2)).
- *Concentration Levels::* Unconcentrated (<1500) vs. High (>2500).
- *Result::* Search HHI > 7300, Mobile OS > 5800, Ride-Hailing ~6150.

AUDITED CASE CONCLUSION

Market tipping, HHI concentration analysis, switching frictions, and Uber China vs. Didi.

Executive Dilemma: Executive Lab

Framing complex management dilemmas and evaluating corporate trade-offs.

EXECUTIVE CHALLENGE

Strategic Tension

Corporate Strategic Tension

- **Margin Divergence::** Why **G** Google makes 30%+ margin while ^{Uber} Uber struggled.
- **Multi-Homing Trap::** High concentration with zero switching barriers.
- **Defensible Moats::** Three design choices creating genuine user lock-in.
- **Executive Memo::** Assessing platform pricing power beyond market share.

EVALUATION RUBRIC

Requirements

Board-Level Deliverable

- **Actionable Recommendation:** Formulate a concrete 3-bullet decision brief for senior leadership.
- **Quantitative Defense:** Support recommendations using audited financial metrics from the lab spreadsheet.
- **Risk Mitigation:** Address channel friction, partner resistance, or competitive retaliation.
- **Implementation Timeline:** Define immediate 90-day execution milestones.

Lab Transition: 120-Minute Hands-On Execution

Transitioning from lecture observation to hands-on data manipulation and strategic decision-making.

EXERCISE 1: FOUNDATIONAL PRACTICE

60 Min Lab

Applied Excel Lab

Hands-on spreadsheet protocol using audited historical numbers:

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EXERCISE 2+: ADVANCED PROBLEM SOLVING

60 Min Memo

Executive Lab

Applied scenario analysis, trade-off evaluation, and executive brief:

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